How Much Shorter Can Prediction-Powered Confidence Intervals Get? Headroom for the Mean When Labels, Unlabeled Data and Predictor Are Fixed

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Abstract

Prediction-powered inference (PPI) and its power-tuned variant PPI++ build valid confidence intervals from a small labeled sample, a large unlabeled sample and a fixed pretrained predictor. We ask how much shorter such intervals can be made when none of these three inputs may change. For the population mean we show that, among estimators that debias the predictor with any function of its output, PPI++ is optimal if and only if $\mathbb{E}[Y \mid f]$ is affine in the prediction f; PPI++ captures exactly a fraction $\rho^2 = \mathrm{Corr}(\mathbb{E}[Y|f], f)^2$ of the variance reduction that f can provide, and the remaining headroom has a closed form that can be estimated from data. We test nonlinear control variates (a natural-spline basis and cross-fitted isotonic regression) against classical, PPI, PPI++ and post-stratified intervals on nine real datasets (2000 resamples per setting) and eleven synthetic scenarios with known truth. The answer is conditional. On the real datasets the nonlinear controls shorten intervals by 0.6–1.2% at $n = 200$ and at most 0.6% for $n \geq 500$ relative to a PPI++ without the $[0, 1]$ clipping of the tuning parameter, in line with a theoretical headroom below 2%; larger differences to the reference `ppi_py` implementation (up to 7% on gene expression) are due to that clipping. On synthetic miscalibrated or non-monotone predictors, and on real outcomes whose predictor scores are put on a different monotone scale, widths shrink by 7–27% (up to 35% on re-scaled real scores) as predicted. At small n the richer controls undercover with plain Wald intervals, which a degrees-of-freedom correction largely repairs, and under distribution shift a tuned $\lambda < 1$ loses coverage that $\lambda = 1$ keeps.

1 Introduction

Many scientific analyses have three ingredients: a small sample with gold-standard labels, a large sample with features only, and a pretrained predictor that cannot be modified. Prediction-powered inference [Angelopoulos et al., 2023a] and PPI++ [Angelopoulos et al., 2023b] turn these into confidence intervals whose coverage does not depend on the predictor being accurate, and whose width shrinks when it is. Follow-up work changes the inputs: it trains the predictor on the labeled data with cross-fitting [Zrnic and Candès, 2023], chooses which points to label [Zrnic and Candès, 2024], or exploits strata [Fisch et al., 2024].

We study a different question, with the inputs frozen: *given the labeled set, the unlabeled set and the predictor, how much shorter can a valid interval for the mean be than the intervals available today, and when?* The question matters because labels are the bottleneck and any width reduction at no extra cost is worth having; it is also easy to answer wrongly, since a narrower interval that undercovers is not a gain.

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Contributions. (i) A short exact analysis (Section 3) of the class of estimators $\hat{\theta}_h = \bar{Y}_L - (\bar{h}_L - \bar{h}_U)$ with arbitrary $h(f)$: the optimal h is a shrunken regression function $\frac{N}{n+N}\mathbb{E}[Y|f]$; PPI++ is optimal iff $\mathbb{E}[Y|f]$ is affine in f; PPI++ realises a fraction ρ^2 of the possible variance reduction over the classical interval; and the reference implementation's clipping of λ costs width when the predictor's scale is small. (ii) Two simple estimators with nonlinear controls and a degrees-of-freedom-corrected variant (Section 4); they are mild extensions of PPI++, not a new inference principle. (iii) An empirical evaluation with repeated sampling, uncertainty on both coverage and width, and a theory-versus-experiment check (Section 5), including negative and failure cases: no gain on real predictors that are already near-affinely calibrated, undercoverage at small n, and fragility of tuned estimators under shift.

2 Related work

PPI [Angelopoulos et al., 2023a] debiases predictions on unlabeled data with a rectifier estimated on labeled data; PPI++ [Angelopoulos et al., 2023b] adds a data-driven tuning parameter λ that interpolates between PPI and the classical estimator and covers means, quantiles and regression. Cross-prediction [Zrnic and Candès, 2023] handles predictors trained on the same labeled data; active inference [Zrnic and Candès, 2024] and stratified PPI [Fisch et al., 2024] change which points are labeled. Our setting differs in that nothing about data collection or the predictor changes. Our nonlinear controls use the cross-fitting idea from double/debiased machine learning [Chernozhukov et al., 2016] for a one-dimensional nuisance $\mathbb{E}[Y|f]$. The control-variate structure used here is the one underlying PPI++ with a vector tuning parameter; what we add is the quantification of what is left after the optimal scalar tuning for the mean. We did not read the full texts of the cited works beyond their abstracts and, for `ppi_py`, its source code; statements about them are limited to what the abstracts and the code state.

3 Setting and headroom analysis

Setting. Labeled pairs $(f_i, Y_i)_{i\le n}$ and unlabeled predictions $(f_j)_{j\le N}$ are i.i.d. from one distribution; $f = f(\text{features})$ is the output of the fixed predictor. The target is $\theta = \mathbb{E}Y$ (chosen because it has a closed-form analysis, is the primary target of all cited methods, and is the simplest quantity on which to test whether any room is left). Write $g(f) = \mathbb{E}[Y|f]$, $v(f) = \mathrm{Var}(Y|f)$. For any h with $\mathbb{E}h(f)^2 < \infty$ define

$$\hat{\theta}_h = \frac{1}{n}\sum_{i\le n}\big(Y_i - h(f_i)\big) + \frac{1}{N}\sum_{j\le N} h(f_j). \tag{1}$$

Classical: $h \equiv 0$; PPI: $h = f$; PPI++: $h = \lambda f$ with tuned λ. (Using the labeled f_i also in the second mean only re-parametrises λ and gives nothing new.)

Proposition 1. *$\hat{\theta}_h$ is unbiased with asymptotic (and, for fixed h, exact) variance $V(h) = \mathrm{Var}(Y - h)/n + \mathrm{Var}(h)/N$. With $c = N/(n+N)$,*

$$V(h) = V_\star + \Big(\frac{1}{n} + \frac{1}{N}\Big)\mathrm{Var}\big(h(f) - c\,g(f)\big), \qquad V_\star = \frac{\mathbb{E}v(f)}{n} + \frac{\mathrm{Var}g(f)}{n+N}.$$

Hence $h_\star = c\,g$ minimises V over all h, with minimum $V_\star$.

Corollary 1 (Headroom beyond PPI++)**.** *Let $\rho = \mathrm{Corr}(g(f), f)$. The best $h = \lambda f$ (PPI++ with unclipped $\lambda^\star = \mathrm{Cov}(g,f)/((1+n/N)\mathrm{Var}f)$) has $V_{\mathrm{lin}} = V_\star + \frac{N}{n(n+N)}\mathrm{Var}(g)(1-\rho^2)$. The classical variance $V_0 = \mathrm{Var}(Y)/n$ satisfies $V_0 - V_\star = \frac{N}{n(n+N)}\mathrm{Var}g$, so PPI++ achieves a fraction ρ^2 of the variance reduction available from f. Equivalently $V_{\mathrm{lin}}/V_\star = 1 + \Gamma$ with $\Gamma = \frac{N\,\mathrm{Var}g\,(1-\rho^2)}{(n+N)\mathbb{E}v + n\mathrm{Var}g}$, and the asymptotic width ratio of the optimal nonlinear control to PPI++ is $(1+\Gamma)^{-1/2}$. In particular no gain is possible iff g is almost surely affine in f (e.g. any predictor taking two values), and the maximal gain is attained when $\rho = 0$.*

Corollary 2 (Clipping)**.** *$V(\lambda f) = V(\lambda^\star f) + (\frac{1}{n}+\frac{1}{N})\mathrm{Var}(f)(\lambda - \lambda^\star)^2$. The reference implementation `ppi_py` (v0.2.3) clips the tuned λ to $[0,1]$ (we read its source); when $\lambda^\star > 1$, e.g. for predictors on a smaller scale than Y, clipping costs the excess above.*

Proofs are in Appendix A. The minimiser $V_\star$ coincides with the asymptotic variance of the unique influence function when labels are missing completely at random with known probability and f is always observed; we use $V_\star$ as the benchmark for "how short can it get" and argue (without a full proof, Appendix A) that it cannot be beaten by regular estimators of this information structure. The corollaries give three sources of possible shortening relative to the `ppi_py` PPI++: (a) removing the clip, (b) nonlinear h, and (c) nothing beyond $V_\star$. They also give a data-based diagnostic: estimate $\mathbb{E}v$, $\mathrm{Var}g$, ρ from the labeled data and compute Γ before changing the estimator. *Finite samples:* the controls must be estimated; with p estimated coefficients the plug-in Wald interval is expected to be too short by a relative amount of order p/n, which is why we also report t-corrected versions.

4 Estimators compared

All intervals are $\hat{\theta} \pm z_{.975}\widehat{\mathrm{se}}$ unless stated. **Classical:** $\bar{Y}_L$ with $s/\sqrt{n}$. **PPI** ($\lambda = 1$) and **PPI++** via `ppi_py` 0.2.3 (Angelopoulos et al., 2024; PPI++ uses its default tuned and clipped λ; our own PPI $\lambda = 1$ implementation agrees with the package, see `code/test_methods.py`). **PPI++ unclipped:** our implementation of the plug-in optimal λ without clipping. **Post-stratified PPI (K=5):** $\lambda = 1$ within five equal-mass strata of the pooled predictions, an adaptation of the stratification idea of Fisch et al. [2024] to simple random labeling (not their allocation scheme); the standard error accounts for the random stratum shares. **NC-spline PPI++:** control $H(f) \in \mathbb{R}^{k-1}$ from a natural cubic spline with k knots at pooled-prediction quantiles (dummy coding if f has $\leq k+2$ distinct values); vector $\lambda = (\hat{\Sigma}_L + \frac{n}{N}\hat{\Sigma}_U)^{-1}\hat{c}$, $\hat{\theta} = \bar{Y} - \lambda^\top(\bar{H}_L - \bar{H}_U)$, $\widehat{\mathrm{se}}^2 = \widehat{\mathrm{Var}}(Y - \lambda^\top H)/n + \lambda^\top\hat{\Sigma}_U\lambda/N$. The span contains f, so it is asymptotically never worse than PPI++. $k = 4$ was the default fixed before seeing real-data results; $k = 3, 6$ are sensitivity checks. **CF-isotonic:** $h = \hat{g}^{(-k)}(f)$ from 5-fold cross-fitted isotonic regression (monotone direction chosen automatically), scalar λ as above (asymptotically $h_\star$ if g is monotone). **t-corrected variants** (classical, PPI++ unclipped, NC-spline k=4): residual variance divided by $n - p - 1$ instead of $n - 1$ and Student-t_{n-p-1} critical value (t_{n-1} for classical).

5 Experiments

Protocol. *Real data:* the nine `ppi-python` example collections (v0.2.3; Angelopoulos et al. 2024) with their stored $(Y, \hat{Y})$ pairs: galaxy spiral-arm presence, forest loss, AlphaFold disorder, gene expression, census income / private insurance / education (each from a pool of $> 3 \times 10^5$ records), plankton and ballots (the stored labeled parts). Each pool is treated as the population; the truth is the pool mean of Y; in each of $R = 2000$ replications n labeled points and $N = 10^4$ unlabeled points are drawn i.i.d. *with replacement* from the pool ($n \in \{50, 100, 200, 500, 1000, 2000\}$; up to 500 for the small pools forest and ballots). This makes the i.i.d. assumption hold exactly for the empirical distribution; it also means for small pools that the unlabeled sample contains repeats. *Synthetic:* $f \sim N(0,1)$, $Y = g(f) + \varepsilon$ with known truth: Hermite scenarios $g = \cos\phi\, f + \sin\phi\,(f^2 - 1)/\sqrt{2}$ ($\phi \in \{0°, 30°, 60°, 90°\}$, $\rho^2 = \cos^2\phi$, $\varepsilon \sim N(0,1)$), lognormal $g \propto e^{\beta f}$ ($\beta \in \{1, 1.5\}$), logistic Bernoulli outcomes $g = \mathrm{expit}(-1 + bf)$ ($b \in \{1, 3\}$), a binary predictor, and two scenarios with f shrunk by 0.25 ($R = 2000$ each, $n \in \{50, \ldots, 1000\}$, $N = 10^4$; $N \in \{200, 10^3, 10^5\}$ at $n = 200$ are in the released CSV). *Metrics:* empirical coverage with Wilson 95% intervals (half-width ≈ 1 point at $R = 2000$); mean width and the *paired* per-replication width ratio to PPI++ (`ppi_py`) with its standard error; theory columns use Corollary 1 (synthetic: exact moments by quadrature; real: $\mathbb{E}v$ from 10-fold cross-fitted gradient boosting on the pool, so the headroom is, if anything, understated). No multiplicity correction is applied: results are descriptive, and claims rest on patterns across many settings rather than on single tests.

Synthetic results: gains exist, as predicted, when g is not affine. Figure 1 and Table 1 show the width of the NC-spline control relative to PPI++. With affine g (Hermite 0°, binary predictor, weakly nonlinear logistic b=1) the ratio is 0.995–1.001, matching the theoretical 1. With nonlinear g it is 0.899 (30°), 0.775 (60°) and 0.727 (90°, a purely quadratic signal on which PPI++ equals the classical interval), against bounds 0.898, 0.762, 0.714; for lognormal and logistic b=3 it is 0.891/0.902/0.933 against bounds 0.845/0.769/0.930. For the heavy exponential g ($\beta = 1.5$) the 4-knot spline recovers less than half of the available reduction (0.902 vs. bound 0.769), while the cross-fitted isotonic control improves with n (0.918 at $n = 200$, 0.866 at $n = 1000$). The dependence on N also follows Corollary 1: for Hermite 60° at $n = 200$ the NC-spline ratio is 0.897, 0.817, 0.775,

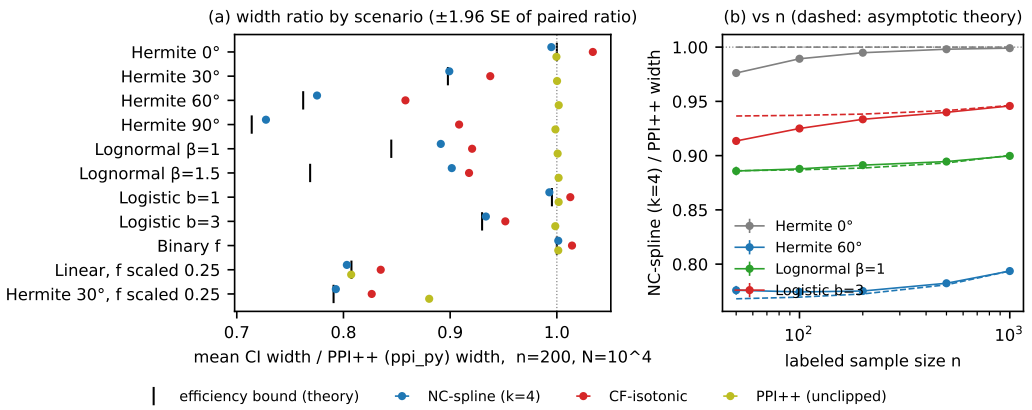


Figure 1: Synthetic scenarios with known truth ($N = 10^4$, $R = 2000$). (a) Mean width relative to PPI++ (`ppi_py`) at $n = 200$; bars ± 1.96 paired SE; black ticks: asymptotic efficiency bound $\sqrt{V_\star / V_{\mathrm{PPI++}}}$. (b) Dependence on n; dashed: asymptotic prediction for the NC-spline ($k = 4$) control. Generated by `code/make_figs.py`.

Table 1: Synthetic scenarios, $n = 200$, $N = 10^4$, $R = 2000$: mean CI width divided by that of PPI++ (`ppi_py`); columns PPI, PPI++ unclip., NC-spline (k=4) and CF-iso are empirical; Bound is the asymptotic efficiency bound $\sqrt{V_\star / V_{\mathrm{PPI++}}}$; last two columns: empirical coverage (nominal 0.95).

Scenario	PPI	PPI++ unclip.	NC-spline	CF-iso	Bound	cov. PPI++	cov. NC-spline
Hermite 0°	1.002	1.000	0.995	1.034	1.000	0.958	0.952
Hermite 30°	1.014	1.000	0.899	0.938	0.898	0.944	0.943
Hermite 60°	1.084	1.002	0.775	0.858	0.762	0.946	0.949
Hermite 90°	1.238	0.999	0.727	0.908	0.714	0.948	0.949
Lognormal β=1	1.035	1.001	0.891	0.921	0.845	0.941	0.944
Lognormal β=1.5	1.108	1.002	0.902	0.918	0.769	0.940	0.936
Logistic b=1	2.213	1.002	0.993	1.013	0.995	0.948	0.946
Logistic b=3	2.140	0.999	0.933	0.952	0.930	0.949	0.950
Binary f	1.137	1.001	1.001	1.014	1.000	0.942	0.942
Linear, f scaled 0.25	1.003	0.807	0.803	0.835	0.807	0.936	0.935
Hermite 30°, f scaled 0.25	1.003	0.880	0.793	0.826	0.791	0.945	0.942

0.770 for $N = 200, 10^3, 10^4, 10^5$ (asymptotic prediction 0.899, 0.815, 0.773, 0.767). Panel (b) shows the asymptotic prediction tracking the empirical ratios. When f is on a smaller scale than Y (last two rows), most or all of the gain over `ppi_py` comes from removing the λ clip: unclipped PPI++ is already 0.807 for the linear case. Coverage of the NC-spline at $n = 200$ is within $[0.935, 0.952]$ in all scenarios. The number of knots matters little (Table 2): $k = 6$ is 1–6 points narrower than $k = 3$ for nonlinear g and up to 1.4 points lower in coverage at $n = 100$; $k = 4$ is a compromise and was not tuned. Across the 11 scenarios $\times$ five n (Table 4) the worst coverage is 0.912 for the NC-spline at $n = 50$ (0.927 with the t correction).

Real datasets: almost nothing left once λ is unclipped. Table 3 and Figure 2 show the real-data widths. The theoretical bound $V_\star$ computed from the pools is within 2% of the unclipped PPI++ width for every dataset at $n = 200$ (largest headroom: forest 0.983, gene expression 0.989, AlphaFold 0.991, others ≥ 0.997), because the stored predictors are nearly affinely calibrated on the pool ($\rho^2 \geq 0.99$ for 8 of 9 datasets, 0.908 for forest; Table 7). Empirically the NC-spline interval is 0.6–1.2% shorter than unclipped PPI++ at $n = 200$ on the seven datasets with continuous scores, and 0.2–0.6% shorter at $n = 500$ (ratios of mean widths), which is what the bound allows; part of the $n = 200$ difference is small-sample optimism of the plug-in standard error (see below). CF-isotonic is 1.5–3.4% *wider* than PPI++ at $n = 200$ on these datasets (gene expression: 2.5% narrower) and converges to it only for $n \gtrsim 1000$, i.e. the price of estimating g nonparametrically exceeds the available gain. Post-stratified PPI is 0.4–1.6% wider than PPI++ at $n = 200$ on these datasets

Table 2: Sensitivity to the number of spline knots k: width ratio to PPI++ (`ppi_py`) at $n = 200$, coverage at $n = 100$ ($N = 10^4$, $R = 2000$); Bound as in Table 1. The last row averages the seven real datasets with continuous scores.

	width ratio, n=200			coverage, n=100			
Setting	k=3	k=4	k=6	k=3	k=4	k=6	Bound
Hermite 0°	0.997	0.995	0.990	0.943	0.939	0.931	1.000
Hermite 30°	0.904	0.899	0.893	0.945	0.940	0.931	0.898
Hermite 60°	0.783	0.775	0.766	0.943	0.939	0.929	0.762
Hermite 90°	0.736	0.727	0.717	0.947	0.945	0.942	0.714
Lognormal β=1	0.916	0.891	0.876	0.934	0.927	0.925	0.845
Lognormal β=1.5	0.935	0.902	0.878	0.941	0.941	0.930	0.769
Logistic b=3	0.977	0.933	0.920	0.931	0.936	0.932	0.930
Binary f	1.001	1.001	1.001	0.945	0.945	0.945	1.000
Real, 7 continuous-score datasets (mean)	0.982	0.979	0.975	0.921	0.918	0.914	–

Table 3: Real datasets: mean CI width relative to PPI++ (`ppi_py`) (ratio of paired widths averaged over $R = 2000$ resamples, $N = 10^4$). Bound: asymptotic efficiency bound at that n. Values below 1 are not gains if coverage is lost; see Table 4.

Dataset	n	Classical	PPI	PPI++ unclip.	Post-strat	NC-spline	CF-iso	Bound
Galaxies	200	1.486	1.002	0.999	1.013	0.992	1.030	0.998
Galaxies	1000	1.414	1.008	1.000	1.008	0.999	1.007	0.998
Forest	200	1.187	1.006	0.999	1.015	0.992	1.015	0.983
Forest	500	1.177	1.006	0.999	1.006	0.996	1.002	0.984
AlphaFold	200	1.724	0.997	0.994	1.006	0.985	1.033	0.991
AlphaFold	1000	1.590	1.002	0.999	1.008	0.997	1.005	0.993
Gene expr.	200	3.145	1.000	0.938	0.951	0.931	0.975	0.930
Gene expr.	1000	2.516	1.000	0.994	1.021	0.992	0.997	0.989
Census income	200	1.258	1.002	0.979	1.004	0.967	1.030	0.979
Census income	1000	1.220	1.000	0.990	0.999	0.986	1.006	0.988
Census health	200	1.408	1.003	1.001	1.016	0.994	1.028	0.999
Census health	1000	1.355	1.006	1.000	1.006	0.999	1.008	1.000
Census educ.	200	1.286	1.004	0.999	1.015	0.993	1.034	0.999
Census educ.	1000	1.249	1.005	1.000	1.007	0.999	1.011	0.999
Plankton	200	1.830	1.032	0.983	0.998	0.983	1.004	1.000
Plankton	1000	1.659	1.075	0.997	1.009	0.997	1.001	1.000
Ballots	200	4.646	1.007	0.991	0.999	0.991	0.999	1.000
Ballots	500	3.524	1.029	0.998	1.014	0.998	1.000	1.000

(gene expression: 4.9% narrower), and its width grows relative to PPI++ with n (e.g. $+6.3\%$ on gene expression at $n = 2000$). Relative to the `ppi_py` PPI++, however, two datasets show visible differences: gene expression (unclipped PPI++ 0.938 of the width at $n = 200$, 0.886 at $n = 50$, 0.971 at $n = 500$) and census income (0.979 at $n = 200$), where $\lambda^\star > 1$ and the clip binds; this is Corollary 2 at work, not nonlinearity (the nonlinear headroom beyond unclipped PPI++ is $\leq 1.2\%$ there). The classical interval is 1.2–$4.6\times$ wider than PPI++ across datasets at $n = 200$, so the benefit of PPI itself is large; the benefit of going further is not. On plankton and ballots the predictor is binary and all of PPI++ unclipped, NC-spline and (nearly) CF-isotonic coincide, as the theory requires.

Validity: the apparent small-n gains are largely not real. Table 4 and Figure 3 summarise coverage. For $n \geq 500$ all intervals are close to nominal (mean 0.947–0.949 over settings), with isolated deviations on the heavy-tailed census income target (0.93–0.94 at $n = 500$–1000). For $n \leq 100$ the plug-in Wald intervals undercover on real data: classical 0.934, PPI++ 0.905 on average, and NC-spline 0.891 (min over all real settings 0.714, on ballots where the predictor almost never disagrees with Y and the residual variance is mostly estimated as zero; every PPI-type method fails there at $n \leq 200$, while the classical interval does not). The t-corrected NC-spline restores some coverage (0.901 real, 0.942 synthetic) at the cost of width: in the no-gain synthetic cases (Hermite $0°$, binary predictor, logistic b=1) it is 1.7–4.2% wider than PPI++ at $n = 50$–100, the same price paid by the corrected PPI++ itself. In short, at small n shorter-looking intervals from richer controls

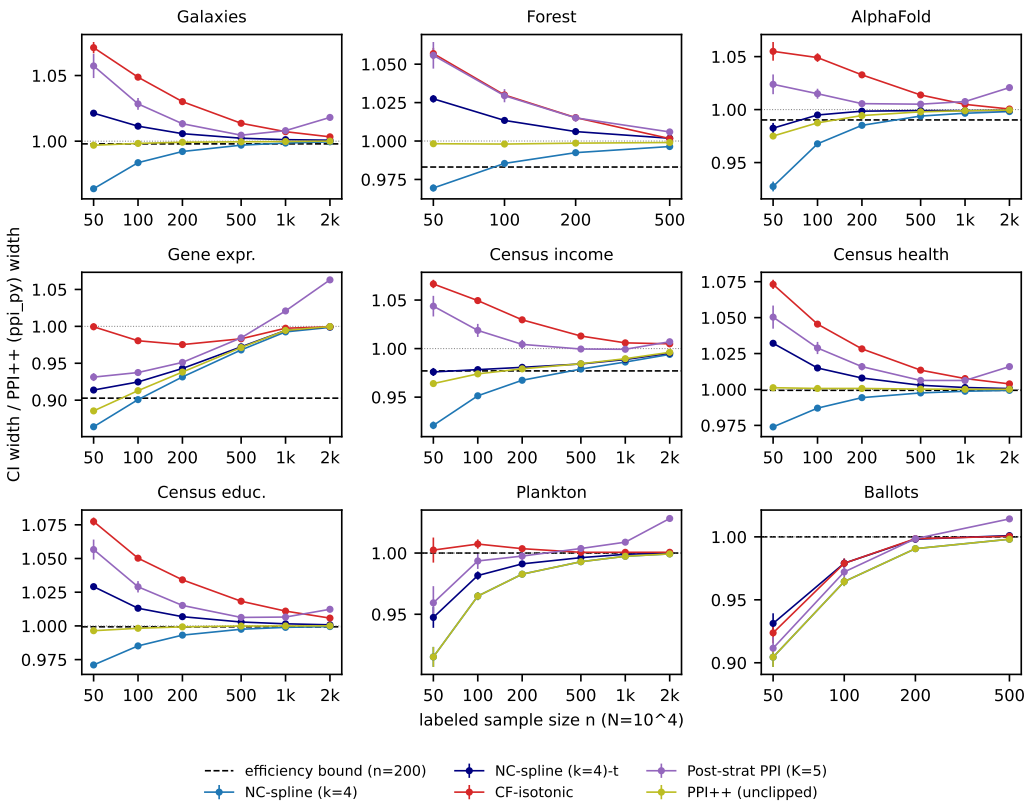


Figure 2: Real datasets: mean width relative to PPI++ (`ppi_py`) vs n ($N = 10^4$, $R = 2000$; ± 1.96 paired SE). Dashed: asymptotic efficiency bound at $n = 200$. Curves of the plain NC-spline below 1 at small n are accompanied by undercoverage (Table 4); the t-corrected variant (dark blue) is wider.

Table 4: Coverage summary: mean empirical coverage over settings with $n \leq 100$ and $n \geq 500$, minimum over all settings, and the number of settings with coverage below 0.93 (real: 9 datasets, 50 settings; synthetic: 55 settings). t: degrees-of-freedom-corrected version.

	Real-data resampling				Synthetic (N=10^4)			
Method	$n \leq 100$	$n \geq 500$	min	#<.93	$n \leq 100$	$n \geq 500$	min	#<.93
Classical	0.934	0.948	0.868	7/50	0.943	0.948	0.926	1/55
Classical-t	0.938	0.948	0.875	5/50	0.947	0.948	0.932	0/55
PPI	0.913	0.949	0.712	7/50	0.946	0.948	0.933	0/55
PPI++ (`ppi_py`)	0.905	0.947	0.796	15/50	0.938	0.947	0.923	3/55
PPI++ (unclipped)	0.904	0.948	0.714	10/50	0.936	0.948	0.915	6/55
PPI++ unclipped-t	0.911	0.948	0.729	8/50	0.943	0.949	0.925	2/55
Post-strat PPI (K=5)	0.908	0.948	0.712	10/50	0.943	0.950	0.932	0/55
NC-spline (k=4)	0.891	0.947	0.714	18/50	0.932	0.949	0.912	9/55
NC-spline (k=4)-t	0.901	0.947	0.729	12/50	0.942	0.949	0.927	1/55
CF-isotonic	0.903	0.948	0.714	11/50	0.939	0.949	0.922	4/55

mostly come with lower coverage, and we found no setting in the real data with $n \leq 100$ in which a richer control gave a gain that survived the coverage check.

Stress test: predictor scores on a different scale. The stored predictors are close to calibrated, which is why Table 3 shows no headroom. To probe the opposite regime on real outcomes we keep each real pool and predictor but change the *scale* on which its output is reported by a fixed monotone map (rank-uniformised score $T(f) = \mathrm{rank}(f)/(\mathrm{pool} + 1)$, or exponential scale $T(f) = \exp((f - \bar{f})/s_f)$); this is a semi-synthetic stress test (7 datasets with continuous scores, $R = 1000$,

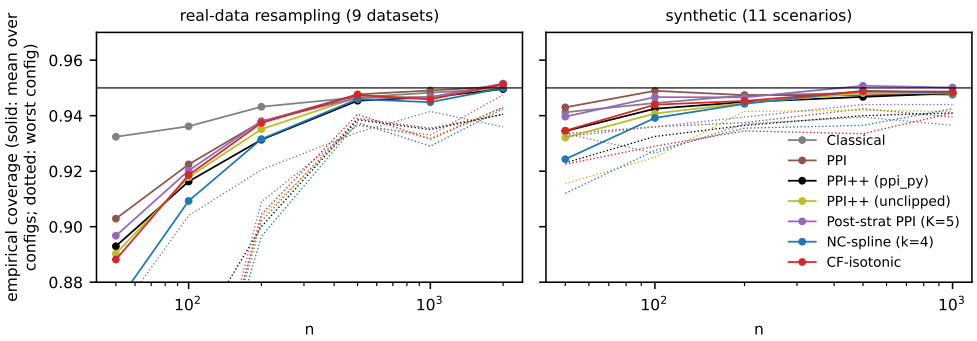


Figure 3: Empirical coverage vs n averaged over settings (solid) and worst setting (dotted).

Table 5: Score-scale stress test at $n = 200$, $N = 10^4$, $R = 1000$: width relative to PPI++ (`ppi_py`) of unclipped PPI++ and the NC-spline, the efficiency bound, and the NC-spline's coverage. $n = 1000$ in `results/distort_summary.csv`.

	rank-uniformised score				exponential score scale			
Dataset	PPI++ unclip.	NC-spline	Bound	cov. NC-s.	PPI++ unclip.	NC-spline	Bound	cov. NC-s.
Galaxies	1.001	0.879	0.881	0.939	0.993	0.945	0.947	0.938
Forest	1.002	0.944	0.922	0.946	0.987	0.966	0.962	0.939
AlphaFold	1.002	0.749	0.720	0.921	0.980	0.916	0.937	0.952
Gene expr.	0.398	0.315	0.317	0.947	0.707	0.462	0.463	0.936
Census income	0.831	0.797	0.797	0.920	0.859	0.768	0.797	0.897
Census health	0.999	0.947	0.948	0.932	1.002	0.963	0.965	0.947
Census educ.	0.841	0.814	0.815	0.942	0.943	0.882	0.879	0.936

$N = 10^4$). At $n = 200$ (Table 5) the NC-spline is 2–35% shorter than unclipped PPI++ (median 6%; largest: gene expression 0.65 on the exponential scale and AlphaFold 0.75 on the rank scale), tracking the bound (e.g. gene expression 0.315 vs. 0.317), and where the λ clip also binds it is up to 54% shorter than `ppi_py` PPI++. Coverage of the NC-spline is 0.897–0.952 (mean 0.941 over 28 settings, lowest on heavy-tailed census income at $n = 200$). Thus nonlinear controls matter when the predictor's output is not on an affine-calibrated scale, but the stored predictors happen not to be of that kind.

Using covariates in addition to the prediction. For the three census collections we also tried a cross-fitted gradient-boosting control on (f, X) with the covariates X shipped with the data ($R = 500$, Table 8). The cross-fitted residual variance on the pool falls by only 0.6%, 0.1% and 0.2% (income, insurance, education) when X is added to f, and the resulting intervals are 1.5–7.7% *wider* than PPI++ (coverage similar), so there is no gain from using X in this way; the cost of fitting a flexible model on $n \leq 2000$ labels exceeds what it can learn.

Distribution shift (negative result). We drew labeled points from the 2013 plankton labels and unlabeled points from the 2014 predictions (truth = mean of the 2014 labels, 0.0714), $N = 10^4$, $R = 2000$. The classical interval, which ignores the unlabeled data, fails as expected (coverage 0.262 at $n = 200$, 0.000 at $n = 1000$). PPI with $\lambda = 1$ stays valid (0.941, 0.940) because the predictor's error is similar in both years; but PPI++ with an unclipped tuned $\lambda < 1$ (identical to the NC-spline on this binary predictor) is shorter and keeps 0.945 at $n = 200$ yet collapses to 0.578 at $n = 1000$ (bias $+0.011$ against a standard error of about 0.006). The `ppi_py` PPI++ has the same coverage and width as PPI here, i.e. its tuned λ was clipped to 1. Tuning λ below 1 trades the calibration-under-shift robustness of PPI for width, so the gain of shortening intervals is conditional on exchangeability, which none of the methods can verify from the data (no-shift control: 0.933–0.954 for all methods, Table 6).

Deployment example. For ballots with the real split ($n = 1027$ labels, $N = 78{,}150$ unlabeled), the classical 95% interval has width 0.0591 while PPI, PPI++ and all our variants have width 0.0131–

0.0133 (estimate 0.6214, $[0.6148, 0.6280]$ for NC-spline); no variant of ours changes it (binary predictor). There is no ground truth for the unlabeled part, so this is illustrative only.

6 Discussion and limitations

Answer to the question. With labels, unlabeled data and predictor fixed, the headroom for the mean beyond a properly tuned (unclipped) PPI++ is $(1+\Gamma)^{-1/2}$ in width and is exactly zero when $\mathbb{E}[Y|f]$ is affine in f. On the nine real collections it is at most about 2% (typically below 1%), so we do not recommend nonlinear controls there; the benefit lies elsewhere (using λ above 1 where needed; PPI itself). On predictors with miscalibrated or non-monotone scores, or scores on an arbitrary scale, the headroom is large (7–27% in our synthetic scenarios) and reachable with a four-knot spline control at $n \gtrsim 200$ without losing coverage.

Limitations. (1) Target: only the mean; quantiles, regression coefficients and other M-estimands are untested, and our theory is for the mean. (2) Real-data experiments resample with replacement from a finite pool and treat its mean as truth, so they test the estimators under i.i.d. sampling from the empirical distribution, not real sampling frames; the plankton and ballots pools are the stored labeled portions only. (3) Headroom for real data uses cross-fitted boosting for $\mathbb{E}v$ and is a plug-in estimate with no confidence interval. (4) Wald-type intervals are used throughout; small-n undercoverage is documented but not solved (ballots, heavy-tailed income); bootstrap or exact methods were not tried. (5) The efficiency statement is argued, not proved, beyond within-class optimality. (6) $k = 4$ and 5-fold cross-fitting were fixed in advance; sensitivity to $k \in \{3, 6\}$ is in the released CSV but not tuned; the spline is clearly too small for heavy exponential g. (7) No multiplicity correction; the conclusions are about consistent patterns, not individual tests. (8) Our extension with covariates uses a boosting model fitted on the labeled data, which departs from a strictly "fixed" predictor. (9) Predictors are those shipped with the data; we did not study predictors of varying quality.

Broader impact. The work is methodological and uses public, de-identified data; the main risk is that shorter intervals presented without a coverage check (as at small n here) overstate certainty.

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A Proofs and derivations

Throughout, $(f_i, Y_i)_{i \le n}$ and $(f_j)_{j \le N}$ are i.i.d. from the same distribution, $g(f) = \mathbb{E}[Y \mid f]$, $v(f) = \mathrm{Var}(Y \mid f)$, and all second moments are finite. For h with $\mathbb{E}h(f)^2 < \infty$ let $\hat{\theta}_h = \frac{1}{n}\sum_{i \le n}(Y_i - h(f_i)) + \frac{1}{N}\sum_{j \le N} h(f_j)$.

Proof of Proposition 1. $\hat{\theta}_h$ is the sum of two independent sample means, hence unbiased for $\theta = \mathbb{E}Y$ with $\mathrm{Var}(\hat{\theta}_h) = V(h) := \mathrm{Var}(Y-h)/n + \mathrm{Var}(h)/N$ exactly, and asymptotically normal as $n, N \to \infty$ by the Lindeberg–Feller CLT applied to each mean (with n/N bounded). Since $Y - g(f)$ is uncorrelated with every square-integrable function of f, $\mathrm{Var}(Y-h) = \mathbb{E}v + \mathrm{Var}(g-h)$. Write $h = cg + \delta$ with $c = N/(n+N)$ (so $1-c = n/(n+N)$). Then $\mathrm{Var}(g-h) = (1-c)^2\mathrm{Var}g - 2(1-c)\mathrm{Cov}(g,\delta) + \mathrm{Var}\delta$ and $\mathrm{Var}(h) = c^2\mathrm{Var}g + 2c\mathrm{Cov}(g,\delta) + \mathrm{Var}\delta$, so

$$V(h) = \frac{\mathbb{E}v}{n} + \Big[\frac{(1-c)^2}{n} + \frac{c^2}{N}\Big]\mathrm{Var}g + 2\Big[\frac{c}{N} - \frac{1-c}{n}\Big]\mathrm{Cov}(g,\delta) + \Big(\frac{1}{n} + \frac{1}{N}\Big)\mathrm{Var}\delta.$$

With $c = N/(n+N)$: $(1-c)^2/n + c^2/N = 1/(n+N)$ and $c/N = (1-c)/n = 1/(n+N)$, so the cross term vanishes and $V(h) = V_\star + (1/n + 1/N)\mathrm{Var}(h - cg)$. □

Proof of Corollary 1. For $h = \lambda f$, $\min_\lambda \mathrm{Var}(\lambda f - cg) = c^2\mathrm{Var}(g)(1-\rho^2)$ with $\rho = \mathrm{Corr}(g(f), f)$ (residual variance of the linear projection of cg on f). By Proposition 1, $V_{\mathrm{lin}} - V_\star = (1/n + 1/N)c^2\mathrm{Var}(g)(1-\rho^2)$, and $(1/n + 1/N)c^2 = \frac{n+N}{nN}\frac{N^2}{(n+N)^2} = \frac{N}{n(n+N)}$. Thus $V_{\mathrm{lin}} - V_\star = \frac{N}{n(n+N)}\mathrm{Var}(g)(1-\rho^2)$. For the classical estimator ($h = 0$), $V_0 = \mathrm{Var}Y/n = (\mathbb{E}v + \mathrm{Var}g)/n$ and $V_0 - V_\star = \mathrm{Var}(g)\big(\frac{1}{n} - \frac{1}{n+N}\big) = \frac{N}{n(n+N)}\mathrm{Var}g$; hence $(V_0 - V_{\mathrm{lin}})/(V_0 - V_\star) = \rho^2$. The width ratio statement follows from width $\propto \sqrt{V}$. For the expression of Γ: $V_{\mathrm{lin}}/V_\star = 1 + \Gamma$ with $\Gamma = \frac{N\mathrm{Var}g\,(1-\rho^2)}{(n+N)\mathbb{E}v + n\mathrm{Var}g}$. □

Proof of Corollary 2. $V(\lambda f)$ is quadratic in λ with leading coefficient $(1/n + 1/N)\mathrm{Var}f$ and minimiser $\lambda^\star = \mathrm{Cov}(g,f)/\big((1+n/N)\mathrm{Var}f\big)$, hence $V(\lambda f) = V(\lambda^\star f) + (1/n + 1/N)\mathrm{Var}(f)(\lambda - \lambda^\star)^2$. Taking $\lambda = \mathrm{clip}(\lambda^\star, 0, 1)$ gives the stated excess when $\lambda^\star > 1$ (or < 0). □

Remark on efficiency (not a full proof). Let $R_k \in \{0,1\}$ indicate labeling, $\pi = n/(n+N)$ known and independent of (f, Y). The observed-data model is then nonparametric for the law of (f, Y), so there is a single influence function, $\varphi = \frac{R}{\pi}(Y - g(f)) + g(f) - \theta$, whose variance divided by $n+N$ is $\frac{\mathbb{E}v}{\pi(n+N)} + \frac{\mathrm{Var}g}{n+N} = V_\star$. The estimator $\hat{\theta}_{cg}$ attains this value, which is why we call $V_\star$ the efficiency bound for this information structure (labels MCAR; f observed everywhere). We use it as the benchmark for the headroom; the claims of the paper do not depend on its optimality beyond the elementary within-class optimality in Proposition 1.

Heuristic for the cost of estimating the controls. With p estimated control coefficients, the classical regression-adjustment literature gives a relative variance inflation of order p/n for the point estimator, and the in-sample residual variance is biased downwards by a factor of order $(n-p-1)/n$; the plug-in Wald interval is therefore too short by about the same relative amount. We do not derive this here; the data in Section 5 are used to check its qualitative implication (undercoverage of the richer control sets at small n, partly repaired by the degrees-of-freedom correction).

B Additional results and reproducibility

Reproducibility. Environment: Python 3.11.15, numpy 2.4.6, scipy 1.17.1, pandas 3.0.6, scikit-learn 1.9.1, matplotlib 3.11.2, ppi-python 0.2.3, joblib 1.6.0. All numbers, tables and figures are produced by scripts in `code/` (see `README.md`): `run_real.py` (Table 3, Fig. 2), `run_synth.py` (Table 1, Fig. 1), `run_extra.py` (shift, ballots, covariates), `run_distort.py` (score-scale stress test), `make_figs.py` (all figures and tables). Seeds are derived from fixed base seeds. A first version of the post-stratified baseline ignored the randomness of the stratum shares and undercovered badly; it was fixed before the reported runs and all reported results come from the fixed code.

Disclosure of LLM use. This study (design, implementation, experiments, analysis and drafting of the manuscript) was carried out by an autonomous LLM-based research agent working unattended; no human edited the results. No LLM is part of the statistical methods studied. The literature was limited to the starter list and the `ppi-python` source; arXiv metadata (titles, authors, dates) of the cited items was verified through the public arXiv API (`code/check_refs.py`); no full texts were read.

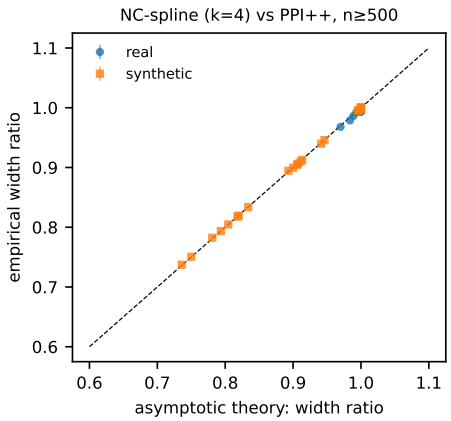


Figure 4: Asymptotic prediction vs empirical width ratio of the NC-spline to PPI++ for $n \geq 500$ (real and synthetic settings).

Table 6: Plankton: distribution shift (labeled 2013, unlabeled and target 2014) and no-shift control; $N = 10^4$, $R = 2000$; the interval is the Wilson interval for the coverage.

Setting	n	Method	Coverage	95% CI of cov.	Width	Bias
shift(2013→2014)	200	Classical	0.262	[0.243, 0.282]	0.0940	+0.0621
shift(2013→2014)	200	PPI	0.941	[0.930, 0.951]	0.0541	-0.0018
shift(2013→2014)	200	PPI++ (`ppi_py`)	0.941	[0.929, 0.950]	0.0540	-0.0018
shift(2013→2014)	200	NC-spline (k=4)	0.945	[0.934, 0.954]	0.0516	+0.0089
shift(2013→2014)	1000	Classical	0.000	[0.000, 0.002]	0.0422	+0.0625
shift(2013→2014)	1000	PPI	0.940	[0.928, 0.949]	0.0258	-0.0014
shift(2013→2014)	1000	PPI++ (`ppi_py`)	0.940	[0.928, 0.949]	0.0258	-0.0014
shift(2013→2014)	1000	NC-spline (k=4)	0.578	[0.556, 0.599]	0.0244	+0.0112
no-shift control(2013)	200	Classical	0.950	[0.939, 0.958]	0.0944	+0.0012
no-shift control(2013)	200	PPI	0.949	[0.938, 0.958]	0.0548	+0.0001
no-shift control(2013)	200	PPI++ (`ppi_py`)	0.933	[0.921, 0.943]	0.0530	-0.0023
no-shift control(2013)	200	NC-spline (k=4)	0.940	[0.928, 0.949]	0.0521	+0.0002
no-shift control(2013)	1000	Classical	0.943	[0.932, 0.952]	0.0421	-0.0003
no-shift control(2013)	1000	PPI	0.954	[0.943, 0.962]	0.0274	+0.0001
no-shift control(2013)	1000	PPI++ (`ppi_py`)	0.950	[0.939, 0.958]	0.0255	-0.0006
no-shift control(2013)	1000	NC-spline (k=4)	0.952	[0.942, 0.961]	0.0254	-0.0001

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Table 7: Pool statistics: $g = \mathbb{E}[Y|f]$ is exact for discrete f and 10-fold cross-fitted boosting otherwise; $\rho^2 = \mathrm{Corr}(g, f)^2$; $\lambda^\star$: unclipped optimal PPI++ coefficient at $n = 200$, $N = 10^4$ (values above 1 are clipped by `ppi_py`).

Dataset	pool	distinct f	$\mathrm{Corr}(Y, f)$	$\mathrm{Var}(g)/\mathrm{Var}(Y)$	ρ^2	$\lambda^\star$ (n=200)
Galaxies	16,743	$> 10^2$	0.741	0.546	0.997	0.95
Forest	1,596	$> 10^2$	0.537	0.276	0.908	0.82
AlphaFold	10,802	$> 10^2$	0.815	0.663	0.990	1.01
Gene expr.	61,150	$> 10^2$	0.963	0.930	0.998	1.13
Census income	380,091	$> 10^2$	0.607	0.372	0.990	1.35
Census health	318,215	$> 10^2$	0.707	0.500	0.999	0.97
Census educ.	318,215	$> 10^2$	0.626	0.392	0.998	0.96
Plankton	421,238	2	0.836	0.699	1.000	0.83
Ballots	1,027	2	0.982	0.963	1.000	0.96

Table 8: Covariate extension ($N = 10^4$, $R = 500$): width ratio to PPI++ (`ppi_py`) of the NC-spline, CF-isotonic and cross-fitted boosting on (f, X) (paired mean $\pm$ SE), and coverage.

Dataset	n	NC-spline	CF-iso	CF-GBM(f, X)	cov. PPI++	cov. CF-GBM
Census income	200	0.967	1.028	1.067±0.002	0.910	0.910
Census income	1000	0.986	1.007	1.023±0.001	0.946	0.944
Census income	2000	0.994	1.006	1.016±0.000	0.954	0.954
Census health	200	0.994	1.027	1.069±0.002	0.970	0.972
Census health	1000	0.999	1.008	1.029±0.000	0.938	0.932
Census health	2000	0.999	1.004	1.015±0.000	0.948	0.942
Census educ.	200	0.993	1.033	1.077±0.002	0.936	0.930
Census educ.	1000	0.999	1.011	1.031±0.001	0.952	0.952
Census educ.	2000	1.000	1.006	1.017±0.000	0.962	0.954

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Table 9: Ballots with the real split ($n = 1027$, $N = 78{,}150$): estimates and 95% intervals (no ground truth for the unlabeled part).

Method	Estimate	Lower	Upper	Width
Classical	0.6280	0.5985	0.6576	0.0591
PPI	0.6211	0.6145	0.6278	0.0133
PPI++ (`ppi_py`)	0.6214	0.6149	0.6280	0.0131
PPI++ (unclipped)	0.6214	0.6148	0.6280	0.0131
Post-strat PPI (K=5)	0.6213	0.6146	0.6280	0.0133
NC-spline (k=4)	0.6214	0.6148	0.6280	0.0131
CF-isotonic	0.6213	0.6147	0.6279	0.0132

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